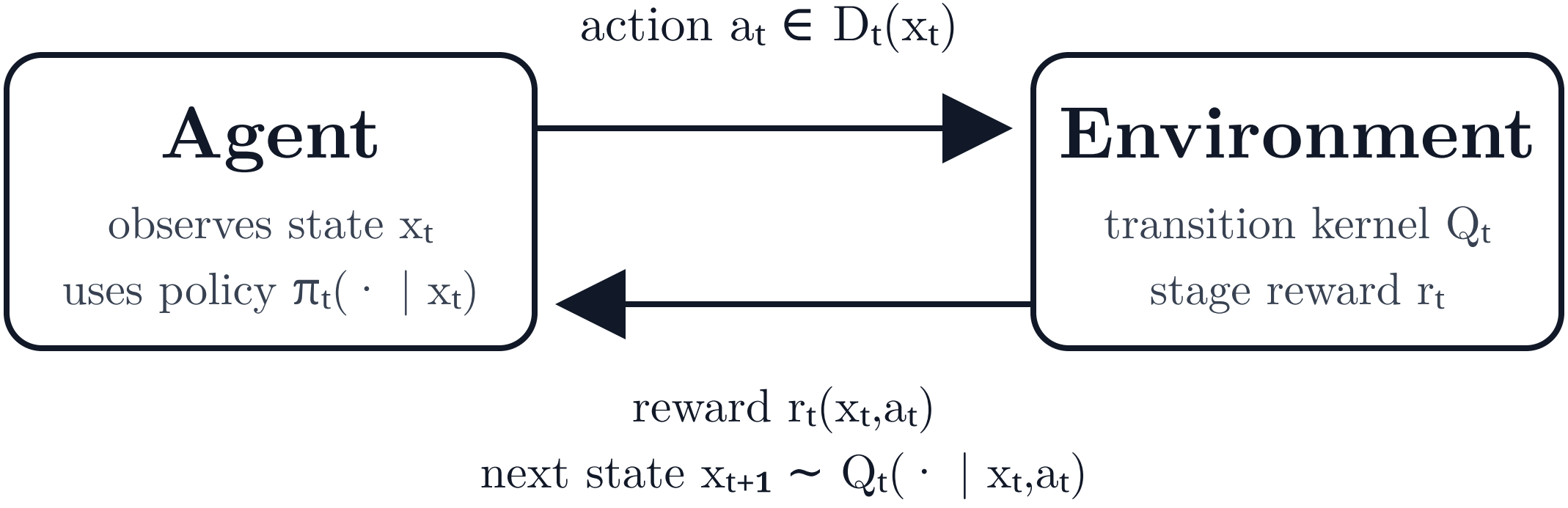
action $a_t \in D_t(x_t)$
Agent
observes state x_t
uses policy $\pi_t(\cdot \mid x_t)$
Environment
transition kernel Q_t
stage reward r_t
reward $r_t(x_t,a_t)$
next state $x_{t+1} \sim Q_t(\cdot \mid x_t,a_t)$